جامعة حفر الباطن
University of Hafr Al Batin

Lecture 6 - Classless Addressing and Subnetting and Supernetting

Objectives

- Classless Addressing.
- Subnetting
- Supernetting

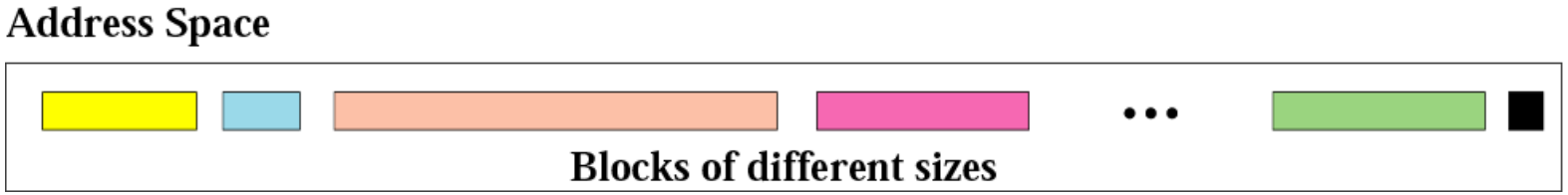
Classless Addressing

Classful vs. Classless Addressing

- Using a **Classful IP addressing** format worked well when the Internet was relatively small.
- But as the number of networks on the Internet grew, the limitations of Classful addresses became apparent.
  - For example, class A address space contains only 125 usable networks in the range 0–127 because networks 0 and 127 are reserved, and network 10 is used for private addressing.

Classful vs. Classless Addressing

- Each of these 125 Class A networks could theoretically contain 16,777,214 hosts.
  - but it's not *realistic* to have more than 16 million hosts on the same network.
- Therefore, in the early 1990s, **the Internet moved away from a classful address space to a classless address space**.
- In other words, the number of bits used for the network portion of an IP address became **variable** instead of **fixed**.



Classful vs. Classless Addressing

- The network portion of **Classful IP addresses** is fixed.
  - Class A addresses use 8 bits.
  - Class B addresses use 16 bits.
  - and Class C addresses use 24 bits.
- A router could determine the address class by inspecting the first byte of the address.
  - A value of 1–128 is Class A, 128–191 is Class B, and 192–223 is Class C.
- For **Classless IP addressing**, there is no longer a relationship between the number of bits used in the network portion and the value of the first byte of the address

Slash Notation

- **How to write the Network Mask?**
  - There is a common way to **write the mask** that is used to **determine the number of bits used for the network component of an IP address.**
  - It is called a **Slash notation**.
  - Slash notation is also called CIDR notation.

- Write / (slash) after the address and then the number of 1 bits in the mask.
  - Class A is /8 (slash 8), Class B is /16 (slash 16), and Class C is /24 (slash 24).

$$A.B.C.D/n$$

Classless Addressing

- **Question:**
  - A small organization is given a block with the beginning address and the prefix length 205.16.37.24/29 (in slash notation). What is the range of the block?
- **Answer:**
  - The beginning address is 205.16.37.24.
  - To find the last address we keep the first 29 bits and change the last 3 bits to 1s.
    - Beginning: 11001111 00010000 00100101 00011000
    - Ending : 11001111 00010000 00100101 00011111
  - There are only 8 addresses in this block.

Classless Addressing

- **Question:**
  - A small organization is given a block with the beginning address and the prefix length 205.16.37.24/29 (in slash notation). What is the range of the block?
- **Answer (Another way):**
  - We can say that the length of the suffix is 32 - 29 or 3.
  - So there are $2^3 = 8$ addresses in this block.
  - If the first address is 205.16.37.24, the last address is 205.16.37.31 (24 + 7 = 31).

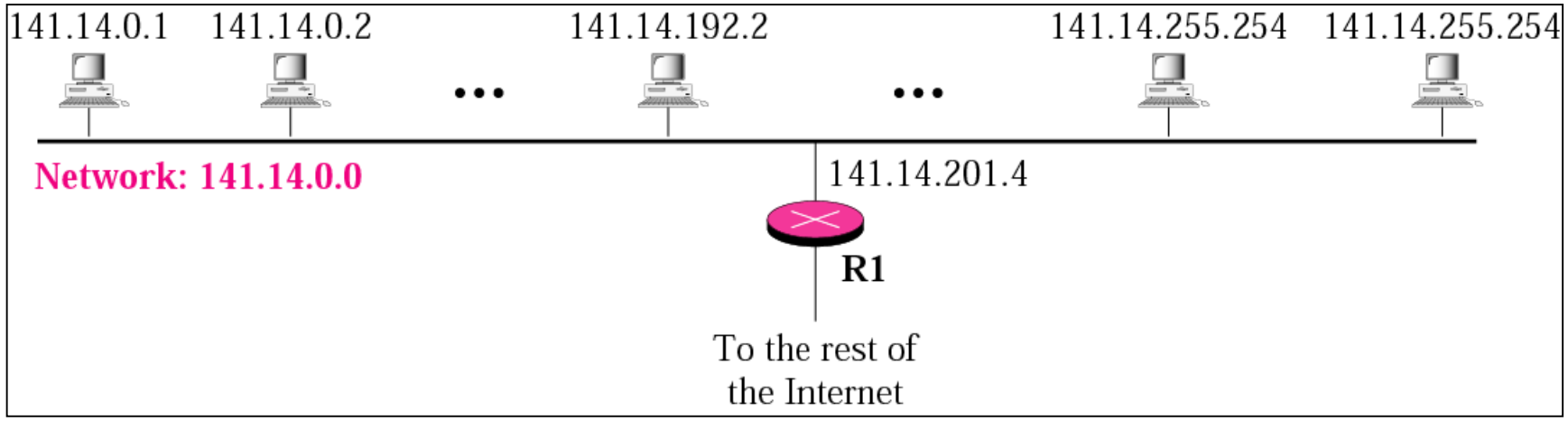
Classless Addressing

- **Question:**
  - What is the network address if one of the addresses is 167.199.170.82/27?
- **Answer:**
  - The prefix length is 27.
  - Which means that we must keep the first 27 bits as is and change the remaining bits (5) to 0s.
  - The 5 bits affect only the last byte.
    - The last byte is 82 or 01010010 in binary.
  - Changing the last 5 bits to 0s, we get 01000000 or 64.
  - The network address is 167.199.170.64/27.

Subnetting

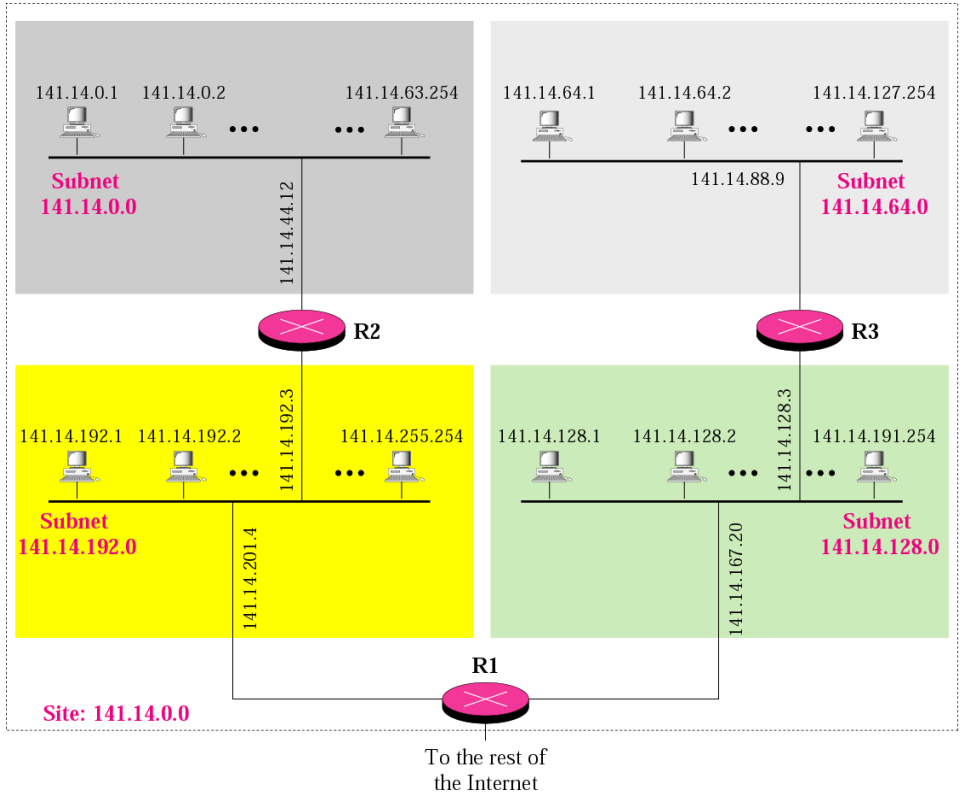
Subnetting

- IP addresses are designed with two levels of hierarchy.



Subnetting

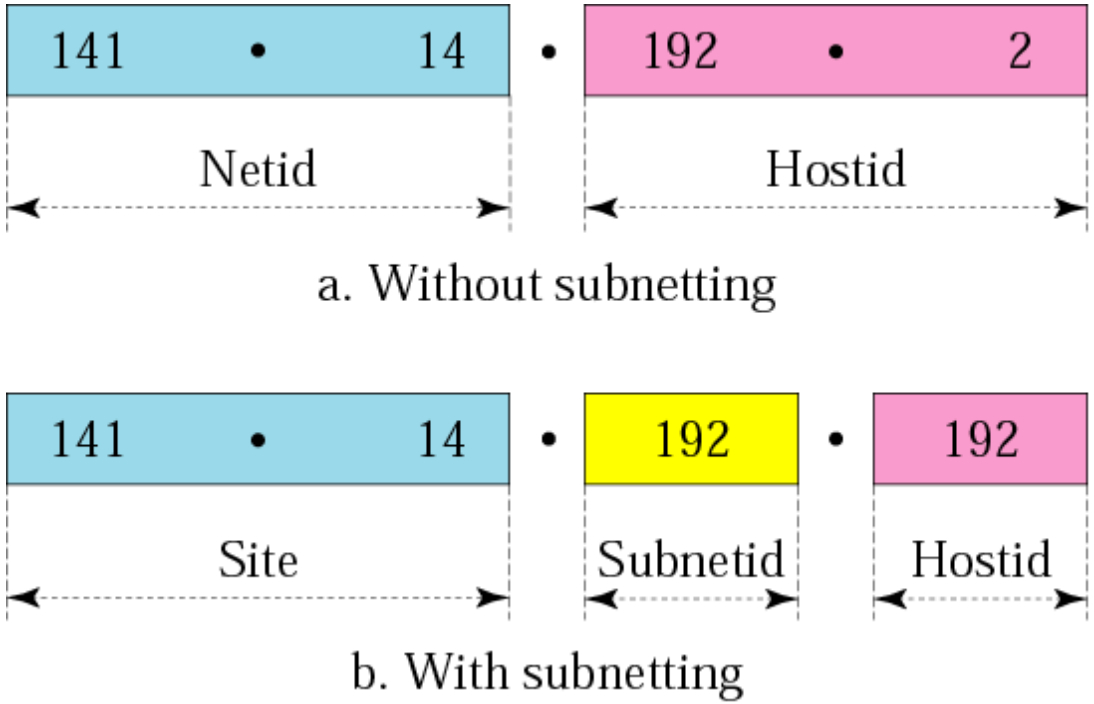
- We can use **Subnetting** to have a network with *three levels* of hierarchy.



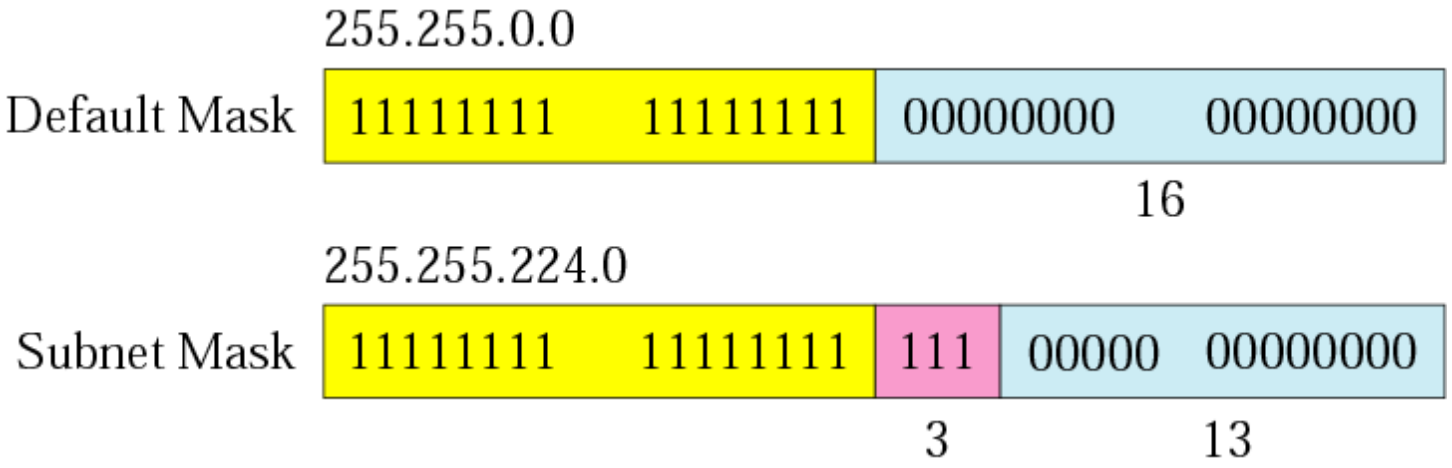
Subnetting

- **A subnetwork (or subnet): is a logical subdivision of an IP network.**
- In subnetting, a single big network is divided into multiple smaller subnetworks.
- The smaller networks are called as **Subnetworks** or **Subnets**.

Addresses in a Network with and without Subnetting

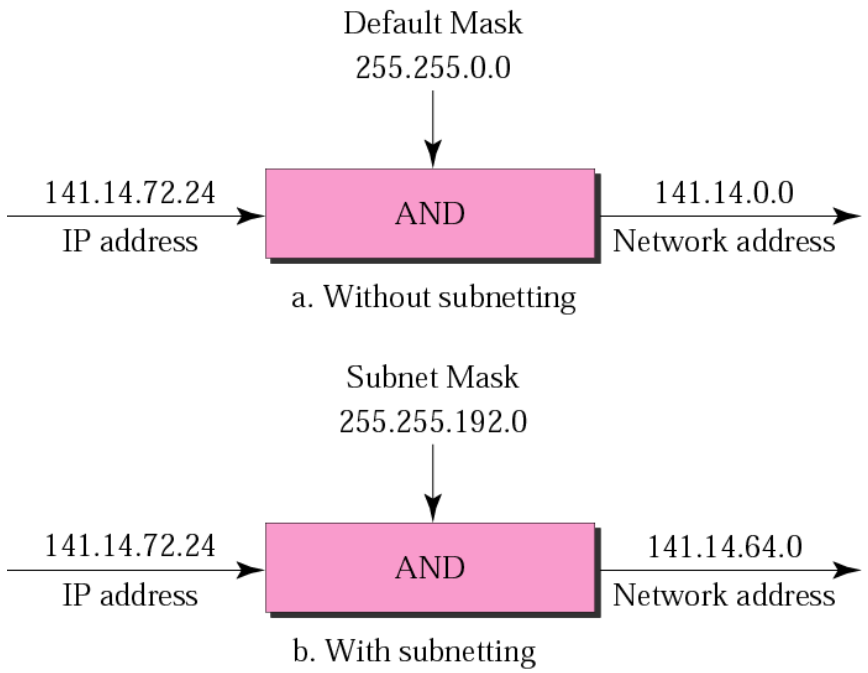


Comparison of a Default Mask and a Subnet Mask



Finding the **Subnet Address**

- Given an IP address, we can find the **subnet address (The address that defines the subnetwork)** the same way we found the network address previously.
  - **We apply the mask to the address.**
  - We can do this in two ways: straight or short-cut.



Finding the **Subnet Address**

- **Straight Method:**
  - In the straight method, we use binary notation for both the address and the mask and then apply the AND operation to find the subnet address.
- **Short-Cut Method:**
  - If the byte in the mask is 255, copy the byte in the address.
  - If the byte in the mask is 0, replace the byte in the address with 0.
  - If the byte in the mask is neither 255 nor 0, we write the mask and the address in binary and apply the AND operation.

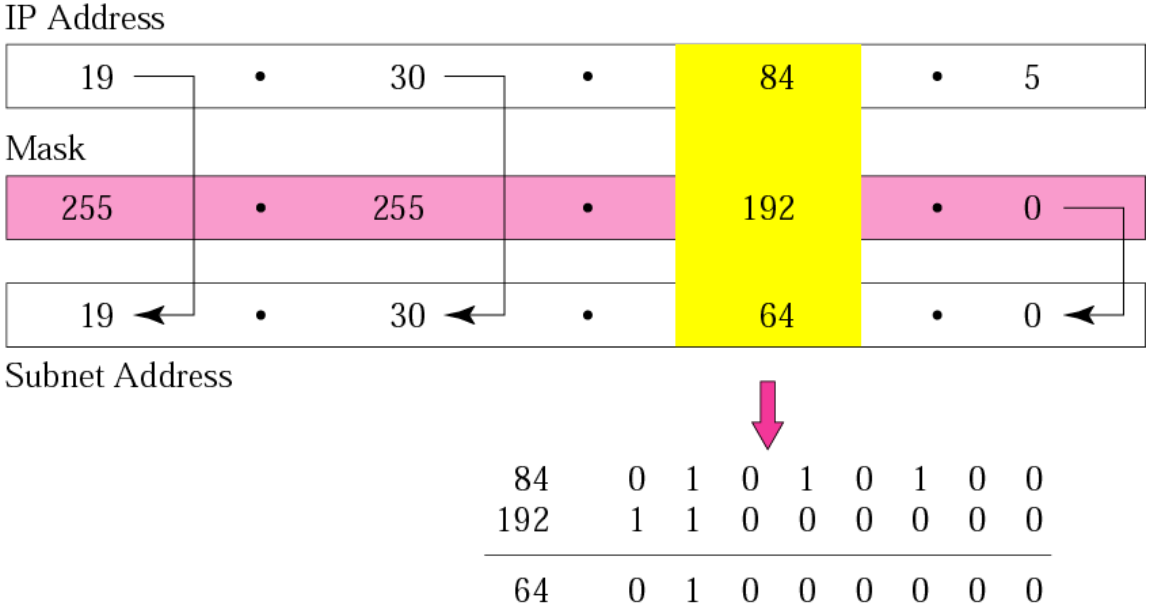
Finding the **Subnet Address**

- **Question:**
  - What is the subnetwork address if the destination address is 200.45.34.56 and the subnet mask is 255.255.240.0?
- **Answer (Straight Method):**
  - 200.45.34.56 → 11001000 00101101 00100010 00111000
  - 255.255.240.0 → 11111111 11111111 11110000 00000000
  - AND → **11001000 00101101 00100000 00000000**

The subnetwork address is: **200.45.32.0**

Finding the **Subnet Address**

- **Question:**
  - What is the subnetwork address if the destination address is 19.30.84.5 and the mask is 255.255.192.0?
- **Answer (Short-Cut Method):**



Designing the **Subnets**

- **Rule #1: The number of subnets must be a power of 2.**

- **Steps:**
  1. Identify the Class and Default Subnet mask of the address.
  2. Find the value of *n* such that 2^n >= *the number of subnets needed*.
  3. Flip n 0's in the *Default mask* to get the *Custom mask*.
  4. Find the number of addresses in each subnet (hint: $2^{\text{the number of 0's}}$)
  5. Calculate the values for each subnet:
     - The subnet address.
     - The first host address
     - The last host address
     - The broadcast address.

Designing the **Subnets**

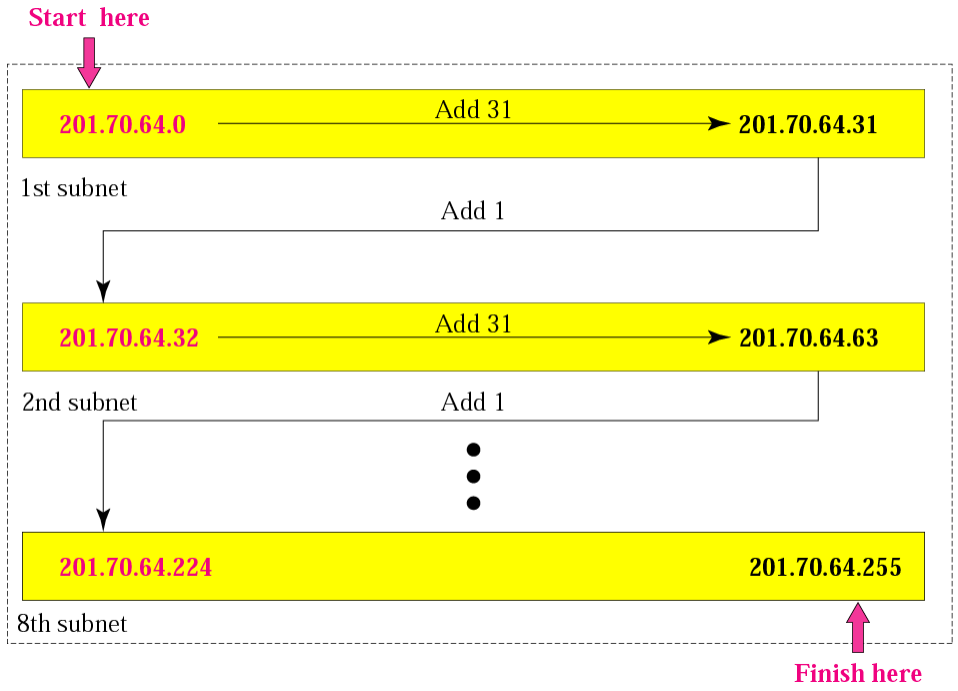
- **Question:**
  - A company is granted the site address 201.70.64.0 (class C). The company needs six subnets. Design the subnets.
- **Answer:**
  - The number of 1s in the default mask is 24 (class C).
  - The company needs 6 subnets. This number 6 is not a power of 2.
  - The next number that is a power of 2 is 8 (2^3).
  - We need 3 more 1s in the subnet mask.
  - The total number of 1s in the subnet mask is 27 (24 + 3).
  - The total number of 0s is 5 (32 - 27).

Designing the **Subnets**

- **Question:**
  - A company is granted the site address 201.70.64.0 (class C). The company needs six subnets. Design the subnets.
- **Answer:**
  - The custom mask is: 11111111 11111111 11111111 **111**00000
  - Or: 255.255.255.224
  - The number of subnets is 8.
  - The number of addresses in each subnet is 2^5 (5 is the number of 0s) or 32.

Designing the **Subnets**

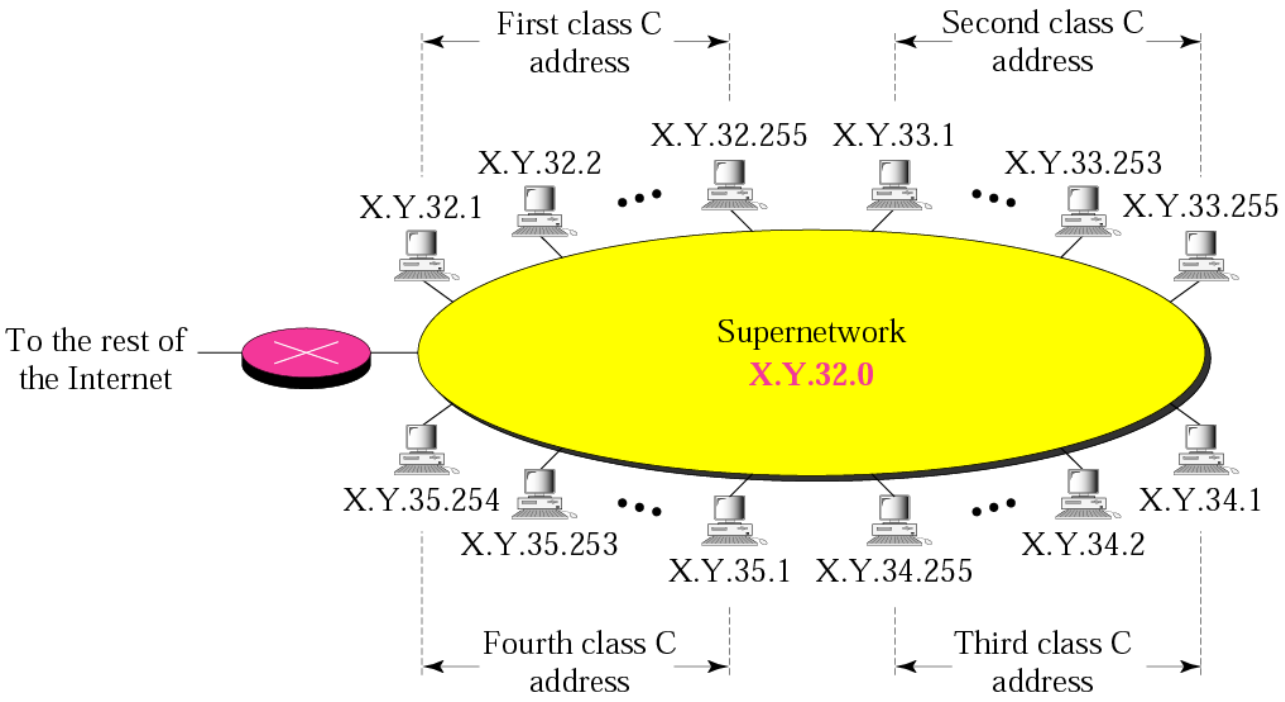
- **Question:**
  - A company is granted the site address 201.70.64.0 (class C). The company needs six subnets. Design the subnets.
- **Answer:**



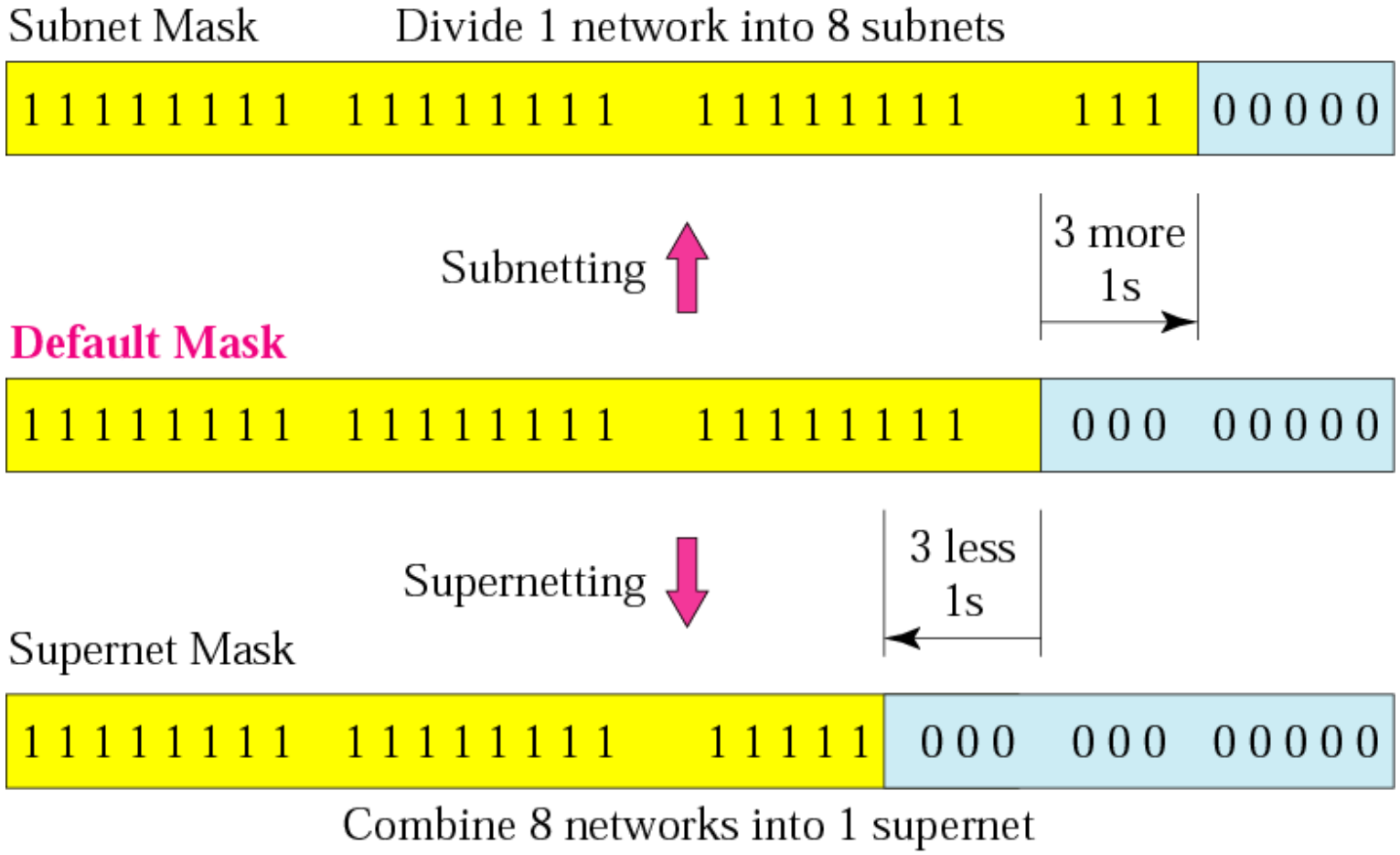
Supernetting

Supernetting

- **Supernetting** is the opposite of Subnetting.
- **In Supernetting, multiple smaller networks are combined into a single big network.**
- The bigger network is called a **Supernetwork** or **Supernet**.



Comparison of Subnet, Default, and Supernet masks



Rules for Supernetworks

- **Supernetwork Rules:**

1. The number of blocks must be a power of 2 (1, 2, 4, 8, 16, . . .).
2. The blocks must be contiguous in the address space (no gaps between the blocks).
3. The third byte of the first address in the superblock must be evenly divisible by the number of blocks.
   - In other words, if the number of blocks is N, the third byte must be divisible by N.

Designing the **Subnets**

- **Question:**
  - A company needs 600 addresses. Can we use the following class C blocks to form a supernet for this company?
    - 198.47.32.0 198.47.33.0 198.47.34.0
- **Answer:**

No, there are only three blocks

The number of blocks must be a power of 2

Designing the **Subnets**

- **Question:**
  - A company needs 600 addresses. Can we use the following class C blocks to form a supernet for this company?
    - 198.47.32.0 198.47.42.0 198.47.52.0 198.47.62.0
- **Answer:**

No, the blocks are not contiguous

The blocks must be contiguous in the address space (no gaps between the blocks).

Designing the **Subnets**

- **Question:**
  - A company needs 600 addresses. Can we use the following class C blocks to form a supernet for this company?
    - 198.47.31.0 198.47.32.0 198.47.33.0 198.47.34.0
- **Answer:**

No, 31 in the first block is not divisible by 4

if the number of blocks is N, the third byte must be divisible by N

Designing the **Subnets**

- **Question:**
  - A company needs 600 addresses. Can we use the following class C blocks to form a supernet for this company?
    - 198.47.32.0 198.47.33.0 198.47.34.0 198.47.35.0
- **Answer:**

Yes, all three requirements are fulfilled:

The number of blocks is a power of 2 (2^2 or 4)

The blocks are contiguous

32 is divisible by 4

Designing the **Subnets**

- **Question:**
  - We need to make a supernetwork out of 16 class C blocks. What is the supernet mask?
- **Answer:**

We need 16 blocks.

For 16 blocks we need to change <u>four</u> 1s to 0s in the default mask.

Default Mask (Class C):	11111111	11111111	11111111	00000000
Supernet mask:	11111111	11111111	1111**0000**	00000000

or

255.255.240.0

Designing the **Subnets**

- **Question:**
  - A supernet has a first address of 205.16.32.0 and a supernet mask of 255.255.248.0. A router receives three packets with the following destination addresses:
    - 205.16.37.44
    - 205.16.42.56
    - 205.17.33.76
  - Which packet belongs to the supernet?

- **Answer:**

  We apply the supernet mask to see if we can find the beginning address.

  205.16.37.44 AND 255.255.248.0 → 205.16.32.0

  205.16.42.56 AND 255.255.248.0 → 205.16.40.0

  205.17.33.76 AND 255.255.248.0 → 205.17.32.0

  Only the first address belongs to this supernet.